Abiodun Olabisi Smart School System Proposal

Title: ***Smart Attendance: An AWS-Powered Solution for Transparent, Scalable, and Fraud-Proof Academic Attendance at the University of Ilorin***

I. INTRODUCTION / OVERVIEW OF THE PROBLEM

Attendance is one of the most important academic requirements at the University of Ilorin, especially under the strict **75% minimum attendance rule** for eligibility to write exams. However, despite its importance, the current attendance-taking process across many departments is **inefficient, unreliable, and vulnerable to fraud**, affecting both students and lecturers.

In several large classes — often with populations ranging from **300 to over 1,000 students** — attendance is taken through a two-step process:

1. A student receives a **manual attendance number** when they enter the class (e.g., "Number 10"),
2. A link is later shared, and students fill in their **number + name + matric number**.

This method creates serious flaws:

1. Attendance impersonation ("Susan attends, Sue fills the form")

Even when a student attends physically, their number can be stolen and filled by someone **not present** in the class.
This fraud results in:

- genuine students losing marks unfairly,
- wrong data being recorded,
- lecturers misled,
- and exam eligibility put at risk.

2. Inaccurate Data for Lecturers

Lecturers cannot verify:

- who filled the form honestly,
- who was physically present,
- or whether numbers were duplicated, shared, or stolen.

This reduces trust in the entire attendance system.

3. Administrative Burden

Large departments spend hours:

- cleaning spreadsheets
- removing duplicates
- correcting impersonated entries

- validating fake records
- dealing with students arguing about attendance errors.

4. Scalability Problems

As the university grows with **multiple faculties and departments**, the existing system cannot scale to thousands of students reliably.
The inefficiency becomes worse during peak periods like:

- GST courses
- large engineering lectures
- combined faculty classes.

5. Portal Limitations

The University portal — although strong — is often overloaded during registration, fees, or examinations.
Connecting attendance directly into the portal risks:

- downtime
- slow responses
- incorrect syncing
- overloading the university servers.

Thus, the university needs a **lightweight, independent, cloud-backed system** that collects accurate data and syncs with the main portal without stressing it.

The Need for a Solution

A campus like the University of Ilorin deserves a system that:

- **ensures fairness**,
- **prevents impersonation**,
- **supports thousands of students**,
- **automates attendance**,
- **reduces staff stress**,
- and **strengthens academic integrity**.

This proposal presents **Smart Attendance**, an AWS-powered, scalable, secure, and fraud-proof attendance management system designed specifically to solve these challenges.

II. PROPOSED SOLUTION & AWS METHODOLOGY

Solution Name:

Smart Attendance (UniAttend) – An AWS-based GPS-verified, fraud-resistant, cloud-synchronized attendance system for Unilorin.

Core Idea

Only a student **physically present** in class should be able to mark attendance — regardless of the manual number system.

With **GPS geofencing**, **device verification**, **offline caching**, and **AWS cloud infrastructure**, Smart Attendance ensures:

- no impersonation
- no duplicate entries
- no post-class form manipulation
- instant verification
- scalability to thousands of users

System Architecture (AWS Components)

1. AWS Amplify (Frontend Hosting & Authentication)

- Hosts the Smart Attendance web app (mobile-friendly).
- Provides secure login with matric number & school email.
- Avoids stressing the school portal by handling authentication independently.

2. Amazon Location Service (GPS + Geofencing)

This service checks:

- whether a student is within **100 meters** of the classroom,
- whether the GPS location is authentic.

Only students physically present can mark attendance.

3. AWS DynamoDB (NoSQL Database)

Stores:

- attendance records,
- session IDs,
- device fingerprints,
- timestamps,

- student metadata.

DynamoDB is:

- lightning fast,
- scalable to millions of transactions,
- ideal for university-wide deployment.

4. AWS Lambda (Serverless Logic)

Handles:

- GPS validation
- session management
- offline-to-online syncing
- duplicate check prevention
- data sanitization
- fraud detection logic

Lambda scales automatically and reduces server cost.

5. Amazon API Gateway

Provides secure API endpoints for:

- marking attendance
- lecturer dashboard
- class session creation
- syncing offline data
- device validation

6. Amazon S3 (Storage)

Stores:

- logs
- attendance exports (CSV, Excel)
- lecturer reports
- backup snapshots

7. AWS CloudWatch

Monitors:

- system health
- suspicious patterns
- performance metrics
- failed sync attempts
- device fraud attempts

Features Summary

Student features:

- GPS-verified attendance
- Offline mode (auto-sync when online)
- One device per matric (fraud-resistant)
- Clean history of all attendance records

Lecturer features:

- Start attendance sessions
- Live student check-ins
- Export attendance to Excel/PDF
- Fraud alert dashboard
- Easy verification of suspicious entries

Admin features:

- Faculty-wide analytics
- API sync to Unilorin Portal
- Storage and backups
- Departmental segmentation

III. AIMS & OBJECTIVES (SMART)

1. Reduce attendance impersonation by 95% within one semester.

By GPS + device binding + dynamic session codes, fraud becomes extremely difficult.

2. Improve attendance accuracy to 99% across participating departments.

Fake entries, double entries, and impersonation are eliminated.

3. Reduce lecturer administrative workload by 40% by automating analysis and reporting.

Cuts time wasted cross-checking forms and cleaning sheets.

4. Ensure the system can serve 10,000–20,000 users with <2 seconds response time.

AWS infrastructure guarantees speed even during peak usage.

5. Make attendance portable across all faculties within 6–12 months.

Scalable architecture supports future expansion.

IV. ANTICIPATED BENEFITS

1. Academic Integrity

- Every student earns attendance fairly.
- No one is cheated due to impersonation.
- Lecturers trust attendance data again.

2. Saves Time for Everyone

- Students sign in instantly — no long queues, no forms.
- Lecturers avoid manual tallying or duplicate corrections.
- Departmental staff get automated reports.

3. Independence from Portal Downtime

Since the system uses AWS infrastructure:

- It works even when the school portal is offline.
- Attendance syncs only when server load is low — no stress on the portal.

4. Scalable to Hundreds of Departments

The AWS serverless design allows the system to support:

- 50 departments
- 100 departments
- even the entire university

without lag or slowdown.

5. Offline Capability

Students with:

- poor network,
- limited data,
- or temporary internet loss

can still mark attendance.
Data syncs automatically later — no stress.

6. Long-Term Vision

The system can evolve into:

- a university-wide identity tool,
- an exam eligibility validator,
- a cloud-based student analytics dashboard,
- or even a national education attendance standard.

V. REFERENCES / SOURCES

- AWS Documentation (Amplify, DynamoDB, Lambda, S3, API Gateway).
- Studies on geolocation-based authentication.
- Articles on attendance fraud and academic system inefficiencies.
- University of Ilorin handbook (75% attendance rule).
- Case studies of digital attendance systems globally.